OUR DAILY BREAD

Acts 2: 21
"And everyone who calls on the name of the Lord will be saved."

MATH PROFICIENCY
(Geometry)

COMMON GEOMETRIC FIGURES:

Line: a collection of points in a straight path that continues infinitely in two directions.

- It takes at least two points to create a line.
- The distance between two points "a" and "b" in the number line is:
  D = A-B/, where A and B are the corresponding values in the number line.
- **Ex:** What is the distance between point A with coordinate -7 from point B with coordinate 2?
  D = /-7 – 2/ = /-9/ = /9/

Line Segment: the part of a line from one endpoint to another.

- If B is between A and C, then AB + BC = AC
- **Ex:** Point B lies on segment AC. AB = 10 and BC = 8; AC = 18.

Parallel Lines: lines that do not meet even when extended infinitely.

Intersecting Lines: lines that meet at one and only one common point.

Perpendicular Lines: intersecting lines that form four right angles.

Collinear Points: three or more points lying in the same single line.

- Two points are always collinear since they always determine a single line.

Plane: a flat surface

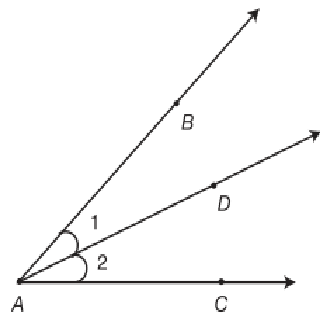
Coplanar: geometrical shapes that lie on the same plane are said to be coplanar.

Skew Lines: lines that are not coplanar.

Ray: half of a line. A ray has one endpoint and continues infinitely in the opposite direction.

ANGLES – formed by two rays and an endpoint or line segments that meet at a point, called the VERTEX.

- **Naming the angles:**
  a) named through the vertex as long as other angle share the same vertex:
  b) for angles with the same vertex, three letters are used, with the vertex always being the middle letter.

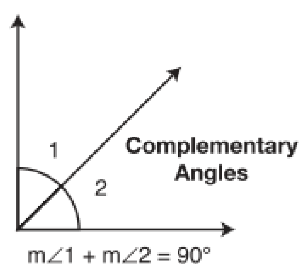


Example:
$\angle$BAC is formed by Ray AB and Ray AC
$\angle$1 can be named as $\angle$BAD or $\angle$DAB

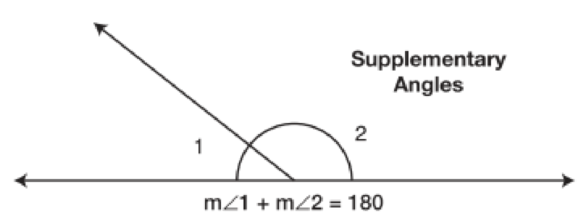
- **Measure of an Angle:** the notation m$\angle$A is used when referring to the measure of an angle and is measured in degrees. Example: if $\angle$1 measures 10^0, then $\angle 1 = 10^0$

CLASSIFYING ANGLES:

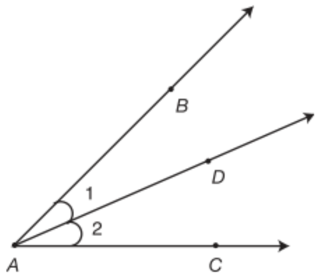
1. **Acute angle –** measures less than 90^0
2. **Right Angle -** measures exactly 90^0
3. **Obtuse Angle –** measures more than 90^0 but less than 180^0
4. **Straight Angle –** measures exactly 180^0 to form a line.
5. **Complementary Angles –** angles whose sum of measures 90^0



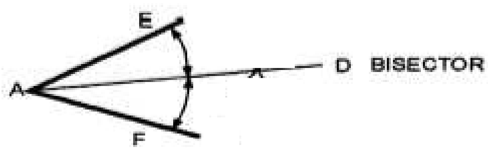
6. **Supplementary Angles –** angles whose sum measures 180^0



7. **Adjacent Angles –** angles which have the same vertex, share one side and do not overlap $\angle 1$ and $\angle 2$ are adjacent angles which share a common vertex A and same side AD
   - The sum of adjacent angles measures up to the bigger angle they make up
     $m\angle 1 + m\angle 2 = m\angle BAC$

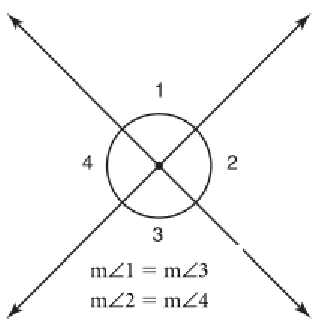


8. **Angle Bisector –** a line which divides an angle into two equal parts $m\angle EAD = m\angle FAD$

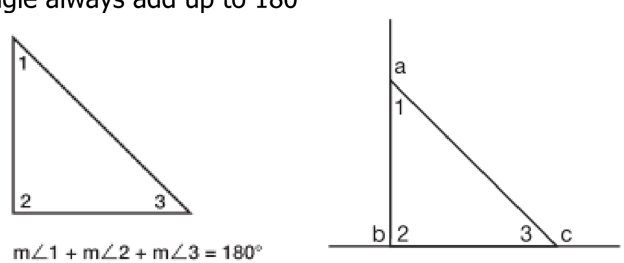


9. **Vertical Angles –** pair of angles found on opposite sides of two intersecting lines.
   $\angle 1$ and $\angle 3$ are vertical angles
   $\angle 2$ and $\angle 4$ are vertical angles

- Vertical angles have equal measures
- Vertical angles are supplementary to adjacent angles
  $m\angle 1 + m\angle 2 = 180$
  $m\angle 3 + m\angle 4 = 180$
- The sum of all adjacent angles around a common vertex is always equal to 360^0



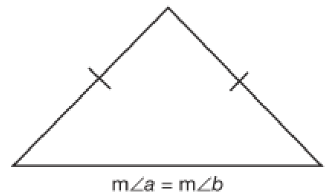
TRIANGLES – the measure of the three angles in a triangle always add up to 180^0



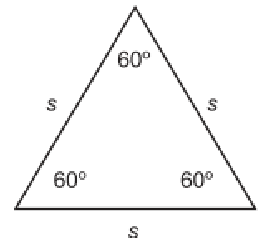
- Triangles have three exterior angles. In the example $\angle a$, $\angle b$ and $\angle c$ are the exterior angles of the triangle.
- An exterior angle is equal to the sum of the non-adjacent interior angles:
  Ex: $m\angle a = m\angle 2 + m\angle 3$
  $m\angle b = m\angle 1 + m\angle 3$

CLASSIFYING TRIANGLES:

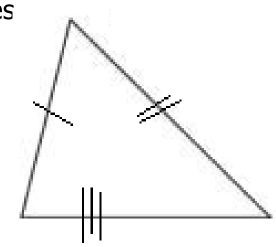
1. **Isosceles triangle –** triangle whose equal angles are opposite equal sides.



2. **Equilateral triangle –** triangle whose all sides are equal to and all angles are 60^0

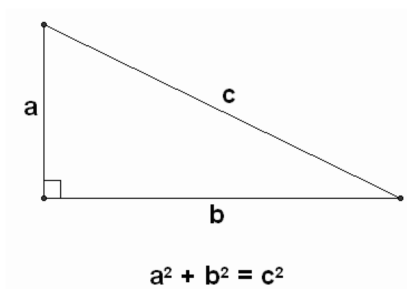


3. **Scalene triangle –** a triangle with all three sides of different measures

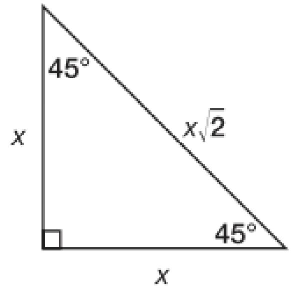


PYTHAGOREAN THEOREM

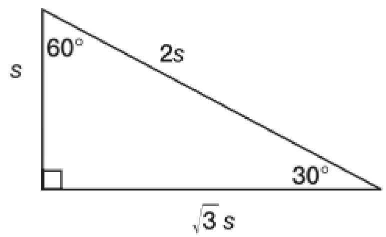
- **Right triangle –** a triangle whose largest angle is 90^0
- Hypotenuse – the side opposite the right angle, also the longest side of a right triangle. The sides other than the hypotenuse are called the legs of the right triangle.
- $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs and c represents the hypotenuse of a right triangle.



- **Isosceles Right triangle –** right triangles with two equal sides, two equal angles and one right angle. The length of the hypotenuse = $\sqrt{2}$ x the length of a leg of the triangle:



- The ratio of 300-600-900 right triangle has a unique ratio of:



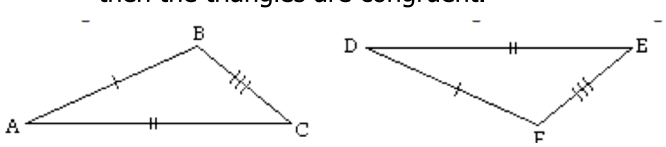
TRIANGLE INEQUALITIES

- The sum of any two sides of a triangle should always be greater than the third side.
- The longest side of a triangle is opposite the smallest angle.

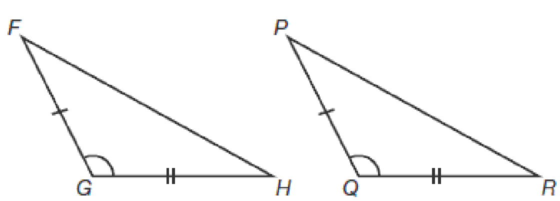
CONGRUENT AND SIMILAR TRIANGLES

Congruent – figures with exactly the same dimensions and shape.

- **S-S-S:** if three sides of one triangle are congruent to three sides of another triangle, then the triangles are congruent.

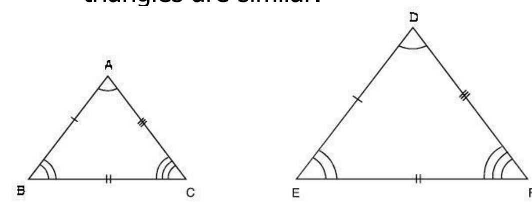


- **S-A-S:** if two sides and the included angle of one triangle are congruent to two sides and the included angle of another triangle, then the triangles are congruent.



- **A-S-A:** if two angles and the included side of one triangle are congruent to two angles and the included side of another triangle, then the triangles are congruent.

- **Similar:** figures with the same shape and whose dimensions are in the same proportion, congruent triangles are also similar.
    - **AA Similarity Postulate:** if two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar.

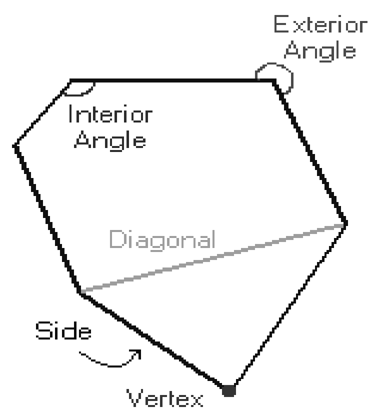


POLYGONS – a closed plane figure made up of several line segments that are joined together. The sides do not cross each other. Exactly two sides meet at every vertex.

Types:

1. **Regular –** all angles are equal and all sides are the same length. Regular polygons are both equiangular and equilateral.
2. **Equiangular –** all angles are equal
3. **Equilateral –** all sides are the same length

Parts:



Side – one of the line segments that make up the polygon.
Diagonal – a line connecting two vertices that is not a side.
Vertex – point where two sides meet. Two or more of these points are called vertices.
Interior Angle – angle formed by two adjacent sides inside the polygon.
Exterior Angle – angle formed by two adjacent sides outside the polygon.

POLYGON FORMULA:

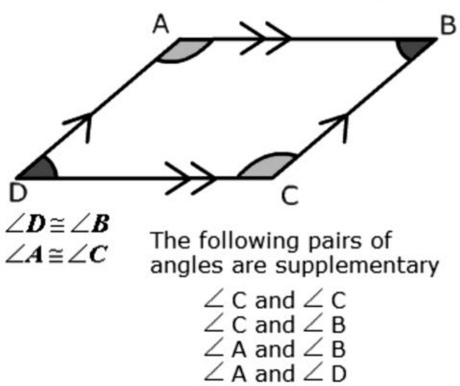
where N = # of sides and S = length from center to a corner

- Sum of the interior angles of a polygon
  = (N - 2) x 180^0
- The number of diagonals in a polygon
  = ½ N (N - 3)
- The number of triangles in a polygon
  = (N - 2)
- Sum (S) of the exterior angles of any polygon
  = 360^0

✓ If two polygons are similar, the ratios of the lengths of corresponding sides are equal and corresponding angles are equal.
✓ If two triangles are similar, then at least two of their corresponding angles are equal.
✓ If two similar polygons have sides in the ratio x:y, then their areas are the ratio x^2:y^2

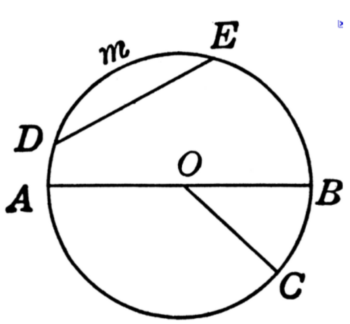
PARALLELOGRAM – a quadrilateral with two pairs of parallel sides. The following are true for parallelograms:

✓ Opposite sides are equal
✓ Opposite angles are equal
✓ Consecutive angles are supplementary
✓ Each diagonal cuts the other diagonal in half



- **Rectangle –** parallelograms with four right angles
- **Rhombus** – parallelogram with four equal sides
- **Square** – parallelogram with four right angles and four equal sides

CIRCLE – closed figure in which each point on the circle is the same distance from a fixed point called the center of the circle.

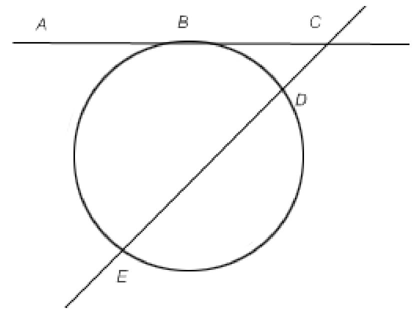


Radius – segment with one endpoint at the center of the circle and the other endpoint on the circle.
Ex: Radius OB and OC

Chord – segment whose endpoint lie on the circle
Ex: Chord DE

Diameter – chord that passes through the center of the circle.
Ex: Diameter AB

Central angles – angles formed by any two radii in a circle whose vertex is the center of the circle.
Ex: ∠BOC



Arc – continuous portion of the circle consisting of two endpoints.
Ex: Minor arc DE (less than a semicircle)
Major arc DCB (more than a semicircle)
Semi-circle AEB (an arc whose endpoints are the endpoints of the diameter of the circle)

Secant – line that contains a chord
Ex: Secant DE

Tangent – line in the same plane as a circle and intersecting the circle at exactly one point
Ex: Tangent AC

Point of Tangency – point where a tangent line intersects a circle
Ex: Point B